



Book I. Prop. XXXIV. Theor. *The opposite sides and angles of any parallelogram are equal, and the diagonal (—) divides it into two equal parts.*

$$\therefore \left\{ \begin{array}{l} \text{blue triangle} = \text{yellow triangle} \\ \text{red triangle} = \text{red triangle} \end{array} \right\} \text{ (pr. I.29)}$$

and — common to the two triangles.

$$\therefore \left\{ \begin{array}{l} \text{orange line} = \text{dashed red line} \\ \text{yellow line} = \text{blue line} \\ \text{black triangle} = \text{black triangle} \end{array} \right\} \text{ (pr. I.26)}$$

and  =  (ax. II).

Therefore the opposite sides and angles of the parallelogram are equal:

and as the triangles  and  are equal in every respect (pr. I.4), the diagonal divides the parallelogram into two equal parts.

Q. E. D.